WhisperVault

Air-Gapped PII Redaction Engine

SNAPDRAGON® AI LAB — BUILD & PRESENT CHALLENGE

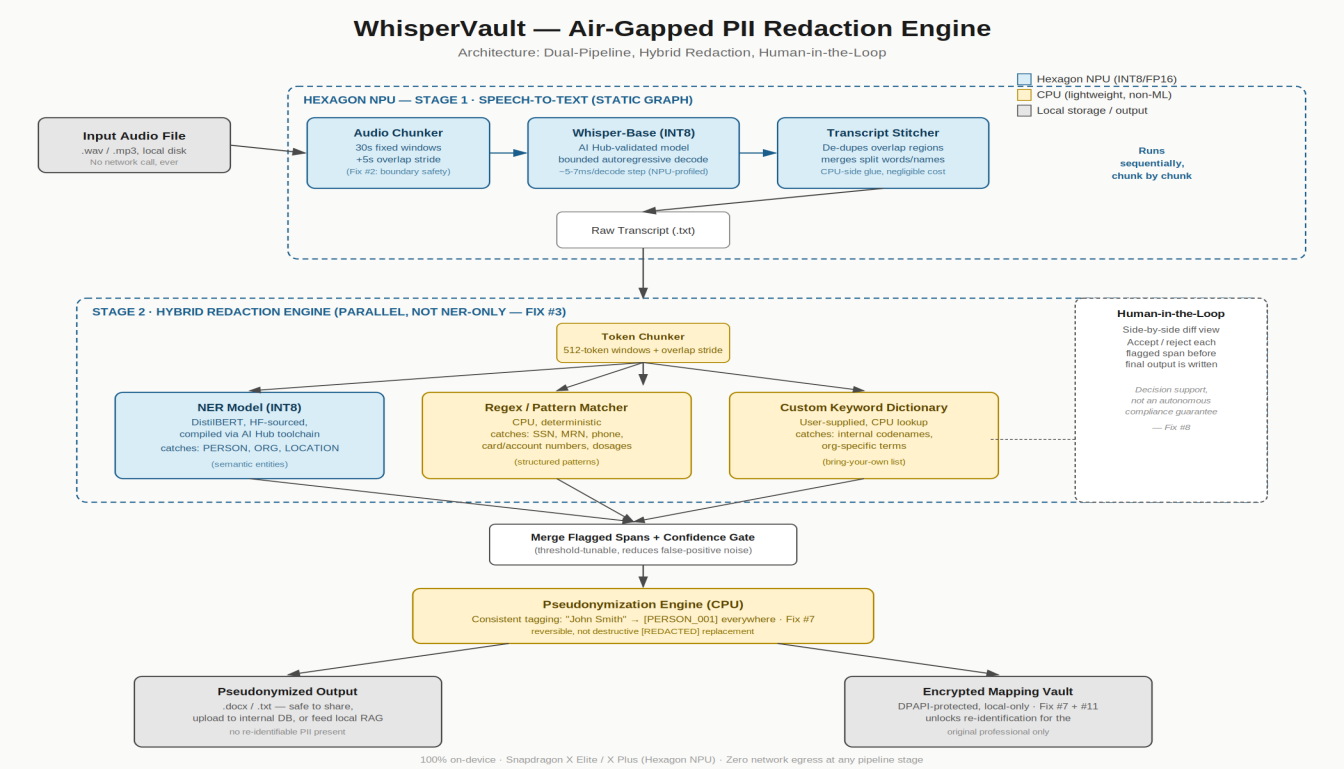
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The Problem: Liability in Plaintext

- Cloud APIs are prohibited by policy & regulations (HIPAA, attorney-client privilege).
- Local transcription leaves behind plaintext PII, making offline files a corporate liability.
- Existing offline tools physically destroy data (irreversible text blanking), preventing legitimate access.

The Solution: WhisperVault

A fully offline, on-device pipeline transcribing and pseudonymizing automatically.



Hexagon Implementation & Hardware Profiling

No Dynamic Overheads: WhisperVault uses strictly static tensors, resolving typical edge-LLM compilation blocks.

Verified Qualcomm AI Hub Profiling Metrics (Job: jpxl8zk8p):

— Target Device: Snapdragon X Elite CRD

— ~44.5ms inference latency per 512-token forward-pass.

— ~168MB peak memory footprint on-device.

DPAPI Vault & Honest Deployment

- **Tri-Layer Accuracy:** Matches Name/Orgs (NPU), standard IDs via Regex (CPU), and Codenames via dictionaries (CPU).
- **Reversible Pseudonymization:** Encrypted tokens map original entities inside the native Windows DPAPI architecture.
- **Zero Deployment Overhead:** Simple Python MSIX file-in/file-out app without virtual driver configurations.
- **Robust Architecture:** 30+ adversarial edge cases tracked and mitigated.